Effect of Interrepetition Rest on Power Output in the Power Clean

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Abstract

Hardee, JP, Triplett, NT, Utter, AC, Zwetsloot, KA, and Mcbride, JM. Effect of interrepetition rest on power output in the power clean. *J Strength Cond Res* 26(4): 883–889, 2012—The effect of interrepetition rest (IRR) periods on power output during performance of multiple sets of power cleans is unknown. It is possible that IRR periods may attenuate the decrease in power output commonly observed within multiple sets. This may be of benefit for maximizing improvements in power with training. This investigation involved 10 college-aged men with proficiency in weightlifting. The subjects performed 3 sets of 6 repetitions of power cleans at 80% of their 1 repetition maximum with 0 (P0), 20 (P20), or 40 seconds (P40) of IRR. Each protocol (P0, P20, P40) was performed in a randomized order on different days each separated by at least 72 hours. The subjects performed the power cleans while standing on a force plate with 2 linear position transducers attached to the bar. Peak power, force, and velocity were obtained for each repetition and set. Peak power significantly decreased by 15.7% during P0 in comparison with a decrease of 5.5% (R1: 4,303 ± 567 W, R6: 4,055 ± 582 W) during P20 and a decrease of 3.3% (R1: 4,549 ± 659 W, R6: 4,363 ± 476 W) during P40. Peak force significantly decreased by 7.3% (R1: 2,861 ± 247 N, R6: 2,657 ± 225 N) during P0 in comparison with a decrease of 2.7% (R1: 2,811 ± 327 N, R6: 2,730 ± 285 N) during P20 and an increase of 0.4% (R1: 2,861 ± 323 N, R6: 2,862 ± 280 N) during P40. Peak velocity significantly decreased by 10.2% (R1: 1.97 ± 0.15 $m \cdot s^{-1}$, R6: 1.79 ± 0.11 $m \cdot s^{-1}$) during P0 in comparison with a decrease of 3.8% (R1: 1.89 ± 0.13 $m \cdot s^{-1}$, R6: 1.82 ± 0.12 $m \cdot s^{-1}$) during P20 and a decrease of 1.7% (R1: 1.93 ± 0.17 $m \cdot s^{-1}$, R6: 1.89 ± 0.14 $m \cdot s^{-1}$) during P40. The results demonstrate that IRR periods allow for the maintenance of power in the power clean during a multiple set exercise protocol and that this may have implications for improved training adaptations.

Key Words force, velocity, fatigue, weightlifting

Introduction

The ability to generate power is one of the main determinants of performance in sports requiring high forces over a short time period (1,2,10,20,21). Several investigations have indicated that improvement in power and athletic performance may be influenced by using weightlifting movements such as the power clean during training (14,25). It has been suggested that fatigue may reduce the effectiveness of power development through decreases in movement velocity and manipulations to exercise technique (23). It is possible that interrepetition rest (IRR) periods may attenuate these unwanted consequences of fatigue during training.

It is clear that muscular power output is reduced with fatigue (5,16), especially during high-intensity activities that require high rates of muscular contraction (24). Reductions in power output result from a decline in both force and velocity (8,9). Power, force, and velocity have been shown to decrease with each repetition during resistance exercise (15,19). Izquierdo et al. (15) demonstrated significant decreases in repetition velocity at one-third (13%) and one-half (8%) of repetitions to failure in the bench press and squat exercises, respectively. Lawton et al. (19) demonstrated a near-linear decrease in power output during a 6 repetition maximum (RM) bench press exercise. Significant decreases in power output were found with each repetition (7.6 ± 9.3%, 17.9 ± 8.1%, 30.3 ± 9.4%, 41.9 ± 11.6%, and 52.9 ± 11.5%, respectively) (19). Duffey and Challis (7) also demonstrated significant decreases in mean and peak velocity during repetitions to failure in the bench press. Drinkwater et al. (6) also found significant decreases in mean and peak power output during the bench press over multiple sets and repetitions. These studies give an insight into the effect of fatigue on power production within single-set configurations; however, these are not common to muscular power training programs that involve multiple set protocols.

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As previously mentioned, it has been suggested that fatigue may reduce the effectiveness of power development through decreases in movement velocity and manipulations to exercise technique (23). Therefore, methods to minimize fatigue are of interest during training for the development of muscular power. One possible method to decrease fatigue with power training is the use of IRR periods. To date, only 2 studies have examined the effect of IRR periods on resistance exercise performance. Haff et al. (11) demonstrated a cluster set configuration (which used IRR periods) resulted in significantly higher barbell velocities in comparison with a traditional set (which did not use IRR periods) in the clean pull when examining sets consisting of both 90 and 120% of 1RM. Additionally, the cluster set resulted in significantly higher vertical barbell displacement when compared with the traditional set at 120% of 1RM (11). Lawton et al. (19) demonstrated greater power output per repetition in the bench press exercise with the use of IRR periods when compared with continuous repetitions. Furthermore, the IRR protocols significantly increased total power output by 21–25% when compared with the continuous set configuration (19). These studies give insight into the use of IRR periods for the maintenance of kinetic and kinematic variables during resistance exercise; however data are limited in that these studies have only observed changes during a single-set protocol and with the clean pull and bench press exercises. Currently, the effects of IRR over multiple sets and repetitions are unknown.

Data would indicate that multiple set strength and power oriented training programs are likely superior to a single set of training for adaptations in muscular strength and power (18). Furthermore, Kaneko et al. (17) demonstrated that the development of muscular power is greatest when training at the intensity at which power is maximized (17). Therefore, the ability to maintain power output across multiple repetitions and sets is of great interest. To date, no research exists examining the combined effects of a constant interset rest period and variable IRR periods in multiple sets on muscular power. This lack of research demonstrates the need to identify the specific interactions between interset and IRR periods for the maintenance of kinetic and kinematic variables during resistance exercise. Understanding how each variable (i.e., power, force, and velocity) is affected over multiple sets during resistance training will lead to further advances in exercise prescription for the development of muscular power. Therefore, the purpose of this investigation was to examine the effect of IRR periods on peak power, force and velocity in the power clean during a multiple set exercise protocol.

Methods

Experimental Approach to the Problem

All subjects participated in 4 testing sessions over a period of 2 weeks. Testing was completed at the same time each day with 72 hours given between sessions. Session 1 consisted of documentation and determination of a 1RM in the power clean. In a randomized order, sessions 2–4 involved subjects performing 3 sets of 6 repetitions at 80% of 1RM with 0 (P0), 20 (P20), or 40 seconds (P40) IRR with 3 minutes of rest given between sets. Peak power, force, and velocity were collected during each protocol for each repetition.

Subjects

Ten male, recreational weightlifters participated in this study (age = 23.6 ± 0.4 years; mass = 80.4 ± 0.9 kg; height = 177.0 ± 0.5 cm; power clean 1RM/mass = 1.39 ± 0.01). The subjects had at least 4 years of weight training experience and 1 year of weightlifting experience. The subjects were required to display proper technique of the power clean exercise for participation in this study as assessed by a National Strength and Conditioning Association Certified Strength and Conditioning Specialist. During the time of the study, all subjects were training for strength and power development and were not currently competing in any outside sports. All subjects read and signed an informed consent approved by the Institutional Review Board at Appalachian State University. No injuries were acquired through participation in this study.

Preliminary Testing: Session 1

All subjects reported to the Neuromuscular and Biomechanics Laboratory for session 1 after refraining from strenuous exercise for a minimum of 48 hours. During this time, the subjects were tested for height, body mass, and a 1RM in the power clean. Power clean 1RM testing was performed as described by Winchester et al. (25). Briefly, the subjects underwent a series of warm-up sets (i.e., percent of 1RM of predetermined 1RM) and several maximal lifts until a 1RM was achieved. All subjects obtained a 1RM greater than their predetermined 1RM.

Protocol Testing: Sessions 2–4

In a randomized order, each subject completed 3 testing sessions over a period of 2 weeks. Testing sessions were separated by a minimum of 72 hours to allow for recovery. The subjects were asked to refrain from strenuous exercise and maintain normal dietary habits between testing sessions. During sessions 2–4, the subjects performed 3 sets of 6 repetitions at 80% of 1RM with either 0 (P0), 20 (P20), or 40 seconds (P40) IRR. Three minutes of rest was given between sets for all conditions. The load of 80% of 1RM was used because it has been shown to be the optimal load for peak and average power in the power clean exercise (4). For each repetition, power cleans were started from the floor. Therefore, during P0, the subjects were asked to bring the bar down in a control manner and reset as quickly as possible. This method eliminates any additional benefits that could have been achieved through the use of the stretch-shortening cycle when lowering and quickly rebounding the barbell off the platform. The subjects were verbally informed of the time remaining during interrepetition and interset rest periods for each protocol. In addition, the subjects were encouraged to give maximal effort with each repetition.

Data Collection

All data was collected and analyzed by the same investigator throughout the study. Kinetic and kinematic data were collected and analyzed as described by Cormie et al. (4). Briefly, testing was conducted with the subjects standing on a force plate (AMTI, BP60011200; Watertown, MA, USA) with 2 linear position transducers (2 LPTs) (Celesco PT5A-15; Chatsworth, CA, USA). The 2 LPTs, one located above-anterior and one above-posterior to the subject, when attached to the bar, resulted in the formation of a triangle, which allowed for the calculation of vertical and horizontal displacements (through trigonometry involving the measurement of displacement and known constants). The combined retraction tension of the LPTs was 16.4 N; this was accounted for in all calculations. Analog signals from the force plate and 2 LPTs were collected at 1,000 Hz using a BNC-2010 interface box with an analog-to-digital card (National Instruments PCI-6014; Austin, TX, USA). The voltage outputs from the force plate and 2 LPTs were converted to force (newtons) and displacement (meters), respectively. LabVIEW (National Instruments, Version 7.1) software was used during data collection and analysis. Peak force was determined from the force-time curve generated from the force plate. Peak velocity was determined from the velocity-time curve generated by using the displacement-time data generated by the 2 LPTs. Power output was calculated by the multiplication of force-time and velocity-time curves. All values where obtained during the second pull of the power clean. This method (4) has displayed intraclass correlation coefficients (ICCs) above the minimum acceptable criterion of 0.70 and were significant at an alpha level of $p \leq 0.05$ (ICC = 0.98). This method has also been validated in comparison with using 1 LPT, 1 LPT + force plate, 2 LPTs, and 2 LPTs + force plate.

Statistical Analyses

A 3 × 3 × 6 repeated measures (protocol × set × repetition) analysis of variance was used to analyze peak values of power, force, and velocity for each repetition during each protocol. When significant values were determined, a Bonferroni post hoc was used to determine statistical

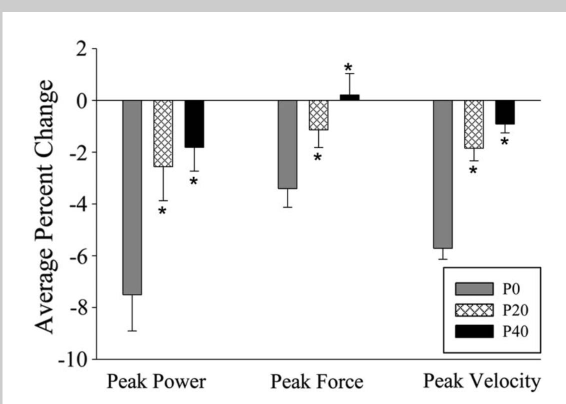

Figure 1. Effect of 0 (P0), 20 (P20), and 40 (P40) seconds of interrepetition rest (IRR) on peak power, force, and velocity. Results are presented as percent change averaged across all repetitions and sets for each protocol (P0, P20, P40). *P20 and P40 significantly different from P0 ($p \leq 0.05$).

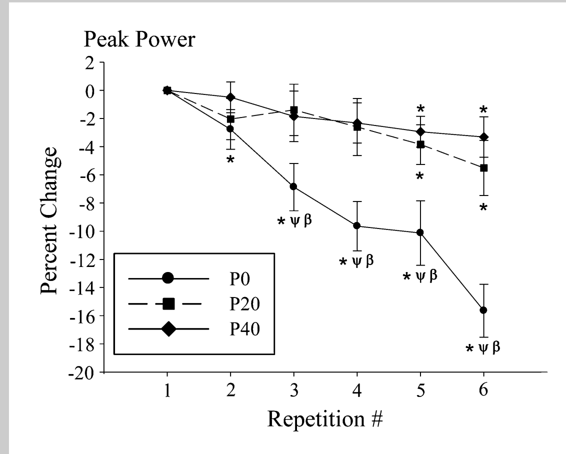

Figure 2. Effect of 0 (P0), 20 (P20), and 40 (P40) seconds of interrepetition rest (IRR) on peak power during each of the 6 repetitions. Results are presented as percent change from the first repetition of each set and averaged across all 3 sets. * = Significantly different from the first repetition. ψ = Significantly different from P20. β = Significantly different from P40 ($p \leq 0.05$).

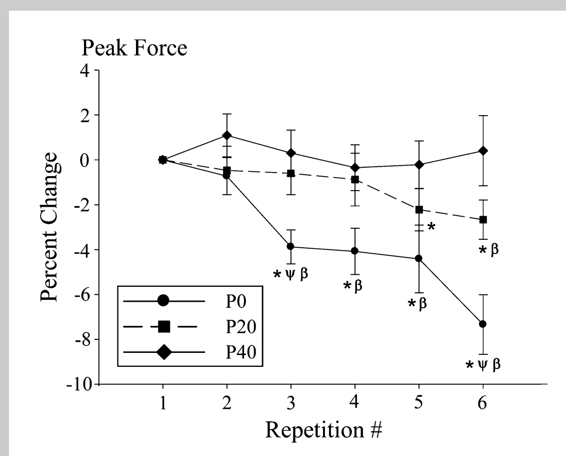

Figure 3. Effect of 0 (P0), 20 (P20), and 40 (P40) seconds of interrepetition rest (IRR) on peak force during each of the 6 repetitions. Results are presented as percent change from the first repetition of each set and averaged across all 3 sets. * = Significantly different from the first repetition. ψ = Significantly different from P20. β = Significantly different from P40 ($p \leq 0.05$).

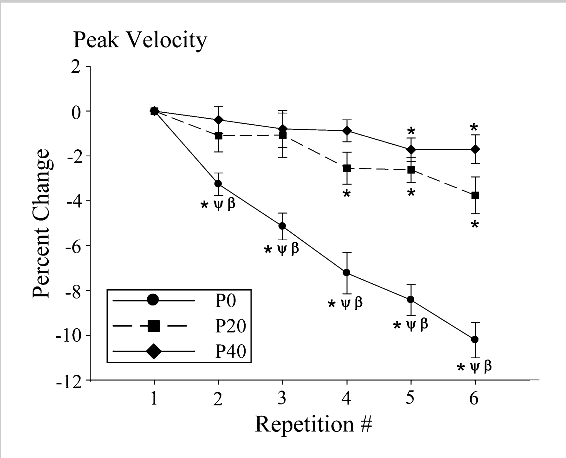

Figure 4. Effect of 0 (P0), 20 (P20), and 40 (P40) seconds of interrepetition rest (IRR) on peak velocity during each of the 6 repetitions. Results are presented as percent change from the first repetition of each set and averaged across all 3 sets. * = Significantly different from the first repetition. ψ = Significantly different from P20. β = Significantly different from P40 ($p \le 0.05$).

significance. The level of significance was chosen at an alpha level of $p \le 0.05$. All data are presented as mean ± standard error. Statistical analysis was performed using SPSS 17.0 (SPSS Inc., Chicago, IL,USA).

Results

The average time to complete each protocol was 430.82 ± 5.06, 725.99 ± 5.13, and 977.44 ± 43.8 seconds (mean ± *SEM*; P0, P20, and P40; respectively). Figure 1 shows the percentage change in peak power, force, and velocity as an average across all sets and repetitions within a given protocol (P0, P20, P40). Percent change in peak power was significantly lower for P0 when compared with P20 and P40 (P0: −7.51 ± 1.39%; P20: −2.56 ± 1.31%; P40: −1.81 ± 0.91%; mean ± *SEM*). Percent change in peak force was significantly lower for P0 when compared with P20 and P40 (P0: −3.41 ± 0.72; P20: −1.14 ± 0.68%; P40: +0.20 ± 0.83%; mean ± *SEM*). Percent change in peak velocity was significantly lower for P0 when compared with P20 and P40 (P0: −5.71 ± 0.42%; P20: −1.85 ± 0.48%; P40: −0.91 ± 0.33%; mean ± *SEM*).

Figures 2–4 show the percentage change in peak power, force, and velocity for repetitions 1–6 averaged across all three sets for each protocol (P0, P20, P40). Peak power significantly decreased by 15.65% (rep 1: 4,563.98 ± 655.08 W, rep 6: 3,881.98 ± 502.08 W) during P0 in comparison with a decrease of 5.50% (rep 1: 4,303.22 ± 566.92 W, rep 6: 4,055.18 ± 581.90 W) during P20 and a decrease of 3.30% (rep 1: 4,549.13 ± 658.52 W, rep 6: 4,363.13 ± 476.01 W) during P40. Peak force significantly decreased by 7.34% (rep 1: 2,861.35 ± 246.76 N, rep 6: 2,656.59 ± 225.21 N) during protocol P0 in comparison to a decrease of 2.67% (rep 1: 2,810.79 ± 326.94 N, rep 6: 2,729.78 ± 284.62 N) during protocol P20 and an increase of 0.40% (rep 1: 2,860.80 ± 322.88 N, rep 6: 2,862.12 ± 280.21 N) during P40. Peak velocity significantly decreased by 10.21% (rep 1: 1.97 ± 0.15 $m \cdot s^{-1}$, rep 6: 1.79 ± 0.11 $m \cdot s^{-1}$) during protocol P0 in comparison with a decrease of 3.76% (rep 1: 1.89 ± 0.13 $m \cdot s^{-1}$, rep 6: 1.82 ± 0.12 $m \cdot s^{-1}$) during P20 and a decrease of 1.70% (rep 1: 1.93 ± 0.17 $m \cdot s^{-1}$, rep 6: 1.89 ± 0.14 $m \cdot s^{-1}$) during P40.

Figures 5–7 show the percent change in peak power, force, and velocity for repetitions 1–6 across each set for each protocol (P0, P20, P40). During set 1, peak power significantly decreased by 13.9% (rep 1: 4,629.94 ± 239.46 W, rep 6: 3,947.09 ± 131.25 W) during P0 in comparison to a decrease of 3.21% (rep 1: 4,354.13 ± 182.46 W, rep 6: 4,210.63 ± 204.97 W) during P20 and a decrease of 4.97% (rep 1: 4,628.87 ± 263.29 W, rep 6: 4,353.80 ± 165.90 W) during P40. During set 2, peak power significantly decreased by 12.76% (rep 1: 4,519.63 ± 291.19 W, rep 6: 3,893.93 ± 207.70 W) during P0 in comparison to a decrease of 7.73% (rep 1: 4,323.59 ± 221.60 W, rep 6: 3,973.12 ±

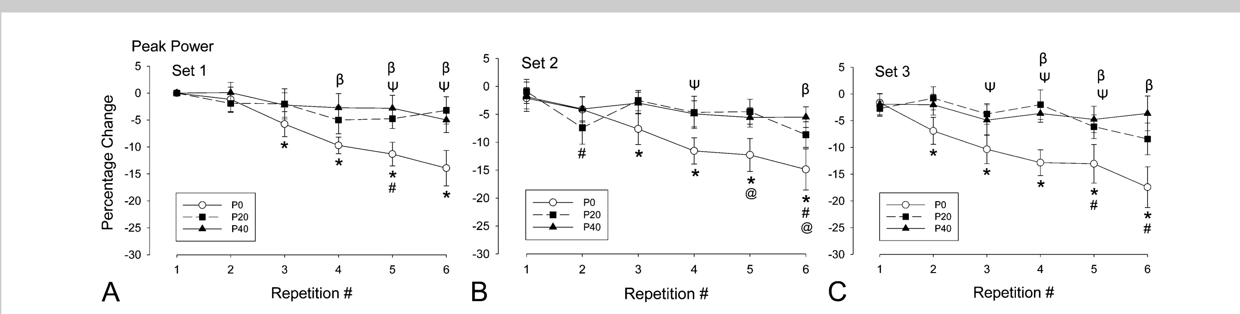

Figure 5. A–C. Effect of 0 (P0), 20 (P20) and 40 (P40) seconds of IRR rest on peak power for each of the six repetitions during each set. Results are presented as percent change from the first repetition of the first set. * = significantly different from the first repetition of P0. # = significantly different from the first repetition of P20. @ = significantly different from the first repetition of P40. y = P0 significantly different from P20. b = P0 significantly different from P40. ¥ = P20 significantly different from P40 ($p \le 0.05$).

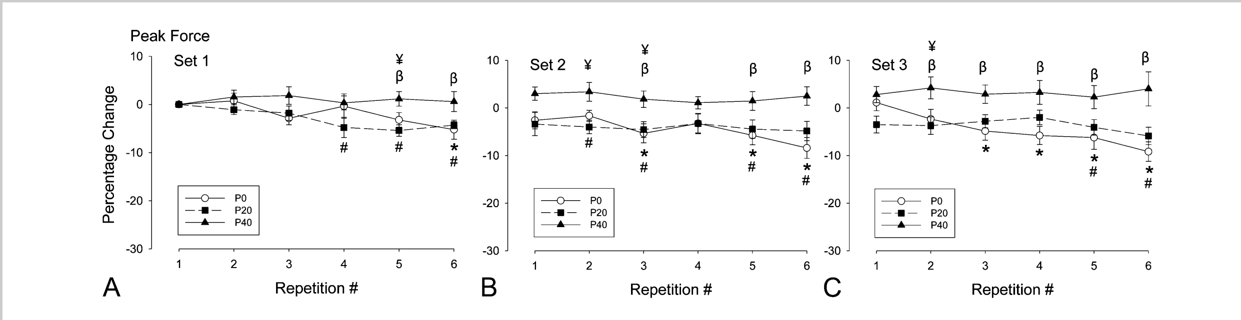

Figure 6. A–C. Effect of 0 (P0), 20 (P20) and 40 (P40) seconds of IRR rest on peak force for each of the six repetitions during each set. Results are presented as percent change from the first repetition of the first set. * = significantly different from the first repetition of P0. # = significantly different from the first repetition of P20. @ = significantly different from the first repetition of P40. y = P0 significantly different from P20. b = P0 significantly different from P40. ¥ = P20 significantly different from P40 ($p \leq 0.05$).

190.10 W) during P20 and a decrease of 3.65% (rep 1: 4,503.95 ± 201.04 W, rep 6: 4,342.53 ± 193.21 W) during P40. During set 3, peak power significantly decreased by 15.81% (rep 1: 4,542.35 ± 212.28 W, rep 6: 3,775.32 ± 211.13 W) during P0 in comparison to a decrease of 5.68% (rep 1: 4,231.95 ± 177.96 W, rep 6: 3,981.78 ± 196.47 W) during P20 and a decrease of 1.71% (rep 1: 4,514.56 ± 210.55 W, rep 6: 4,393.06 ± 126.83 W) during P40.

During set 1, peak force significantly decreased by 5.22% (rep 1: 2,880.13 ± 92.50 N, rep 6: 2,718.53 ± 85.92 N) during P0 in comparison to a decrease of 4.29% (rep 1: 2,874.03 ± 93.55 N, rep 6: 2,750.17 ± 90.29 N) during P20 and an increase of 0.59% (rep 1: 2,807.58 ± 105.78 N, rep 6: 2,820.68 ± 115.04 N) during P40. During set 2, peak force significantly decreased by 5.81% (rep 1: 2,799.91 ± 85.82 N, rep 6: 2,627.27 ± 78.25 N) during P0 in comparison to a decrease of 0.93% (rep 1: 2,780.43 ± 124.23 N, rep 6: 2,732.23 ± 96.27 N) during P20 and a decrease of 0.51% (rep 1: 2,894.91 ± 125.84 N, rep 6: 2,867.57 ± 89.60 N) during P40. During set 3, peak force significantly decreased by 10.3% (rep 1: 2,904.01 ± 71.71 N, rep 6: 2,601.61 ± 55.63 N) during P0 in comparison to a decrease of 2.38% (rep 1: 2,777.91 ± 115.51 N, rep 6: 2,706.93 ± 107.07 N) during P20 and an increase of 1.20% (rep 1: 2,879.92 ± 98.79 N, rep 6: 2,898.10 ± 80.4 N) during P40.

During set 1, peak velocity significantly decreased by 9.14% (rep 1: 2.00 ± 0.05 m·s^{-1}, rep 6: 1.79 ± 0.03 m·s^{-1}) during P0 in comparison to a decrease of 2.19% (rep 1: 1.90 ± 0.04 m·s^{-1}, rep 6: 1.86 ± 0.03 m·s^{-1}) during P20 and a decrease of 1.85% (rep 1: 1.94 ± 0.05 m·s^{-1}, rep 6: 1.91 ± 0.05 m·s^{-1}) during P40. During set 2, peak velocity significantly decreased by 8.58% (rep 1: 1.98 ± 0.04 m·s^{-1}, rep 6: 1.80 ± 0.03 m·s^{-1}) during P0 in comparison to a decrease of 4.76% (rep 1: 1.89 ± 0.05 m·s^{-1}, rep 6: 1.80 ± 0.03 m·s^{-1}) during P20 and a decrease of 1.79% (rep 1: 1.90 ± 0.05 m·s^{-1}, rep 6: 1.87 ± 0.04 m·s^{-1}) during P40. During set 3, peak velocity significantly decreased by 7.64% (rep 1: 1.95 ± 0.05 m·s^{-1}, rep 6: 1.79 ± 0.04 m·s^{-1}) during P0 in comparison to a decrease of 4.42% (rep 1: 1.87 ± 0.04 m·s^{-1},

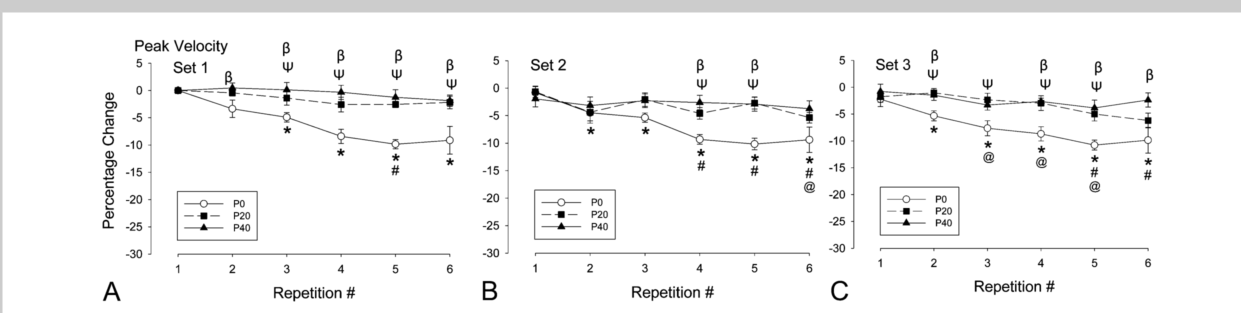

Figure 7. A–C. Effect of 0 (P0), 20 (P20) and 40 (P40) seconds of IRR rest on peak velocity for each of the six repetitions during each set. Results are presented as percent change from the first repetition of the first set. * = significantly different from the first repetition of P0. # = significantly different from the first repetition of P20. @ = significantly different from the first repetition of P40. y = P0 significantly different from P20. b = P0 significantly different from P40. ¥ = P20 significantly different from P40 ($p \leq 0.05$).

rep 6: 1.79 ± 0.04 $m \cdot s^{-1}$) during P20 and a decrease of 1.58% (rep 1: 1.93 ± 0.06 $m \cdot s^{-1}$, rep 6: 1.89 ± 0.04 $m \cdot s^{-1}$) during P40.

Discussion

This study demonstrated that longer IRR periods result in maintenance of peak power, force, and velocity in the power clean during a multiple set exercise protocol. This is in agreement with the findings of previous research by Lawton et al. (19) and Haff et al. (11) demonstrating the effect of IRR periods on exercise performance. Interestingly, this study found significant decreases in power, force, and velocity even within the second repetition of the first set during the P0 condition. This may be due to the inability of the P0 condition to allow for recovery of short-term energy substrates resulting in a rapid rate of fatigue and subsequent decrease in power (11,12,19). The optimization of maximal power output as a stimulus for power improvements with training has been shown by Kaneko et al. (17). Thus, the use of longer IRR periods may be beneficial for increasing power capabilities with training.

Similar to the current investigation, Lawton et al. (19) demonstrated significant decreases in power output during 6 continuous repetitions in the bench press exercise; whereas the IRR protocols (single, doubles, triples) significantly increased total power output by 21–25% compared with the continuous set configuration (19). During the current investigation, P0 displayed significant decreases in power, force, and velocity (15.7, 7.3, and 10.2%, respectively) over 6 repetitions. Haff et al. (11) demonstrated higher barbell velocities with the use of a cluster set (which involved IRR) when compared with a continuous set configuration. This study found significant decreases in power, force, and velocity as early as the second repetition during the P0 set configuration. Izquierdo et al. (15) found significant decreases in velocity at repetitions 3, 4, 5, and 7 (75, 70, 65, and 60% of 1RM, respectively) in the bench press and at repetitions 5, 9, 11, and 15 (75, 70, 65, and 60% of 1RM, respectively) in the squat during continuous repetitions to failure. The authors suggested that to maintain repetition speed that velocity should not decrease more than approximately 13% and 8% from the first repetition in the bench press and squat, respectively. The results from this study demonstrated that power decreased by only approximately 4–6% during the IRR protocols. This indicates that these guidelines can be achieved through the use of IRR over a multiple set and repetition exercise protocol in the power clean.

It is believed that IRR allows for greater power output because of partial recovery of energy substrates and reversal of fatigue (11,12,19). Bogdanis et al. (3) has demonstrated that phosphocreatine (PCr) resynthesis is important for the recovery of power during repeated bouts of sprint exercise. Significant correlations ($r = 0.71 - 0.86$) were found between the resynthesis of PCr and the percentage of restoration of peak power output, peak pedal speed, and mean power during the initial 6 seconds of exercise after 1.5 and 3 minutes recovery (3). Furthermore, Harris et al. (13) demonstrated that PCr resynthesis half-time was calculated to be 21–22 seconds and occlusion of the circulation to a fatigued skeletal muscle inhibits PCr resynthesis. Therefore, it is speculated that IRR of at least 20 seconds may allow for partial PCr resynthesis and maintenance of power, force, and velocity. This notion is supported by Pereira et al. (22) that demonstrated rest interval lengths of 14 to 17 seconds was sufficient to maintain jumping performance during 30 maximal volleyball spikes, whereas a rest interval length of 8 seconds resulted in increased blood lactate concentrations and decreased countermovement jump performance (22). To date, no study has examined PCr resynthesis during an IRR protocol, therefore future research is needed to identify the exact mechanisms relating the maintenance of power production during IRR protocols.

Kaneko et al. (17) demonstrated that the development of muscular power is greatest when training at the intensity at which power is maximized (17). Our data demonstrate that 20–40 seconds of IRR can be useful to maintain power, force, and velocity during a multiple set protocol. Therefore, it could be speculated that this training methodology would induce a greater training stimulus when compared with a continuous set configuration. This may be advantageous for individuals seeking a greater training response with regards to the development of muscular power. Because there were no significant differences between P20 and P40, it can be speculated that there is no additional benefit to resting >20 seconds. In conclusion, 20–40 seconds of IRR allows for minimal decreases in peak power, force, and velocity in a multiple set protocol in the power clean. One mechanism may include the availability of short-term energy systems through the attenuation of fatigue. Training protocols for improvement in power capabilities should consider IRR as a possible method of maintaining power output and thus providing an optimal stimulus for further improvements in power output

Practical Applications

This study demonstrates that using IRR periods is a training method to maintain kinetic and kinematic variables over multiple sets and repetitions during a resistance training session. Therefore, IRR periods should be considered when designing training programs for the development of muscular power. Because there were no significant differences between P20 and P40, the addition of 20 seconds of IRR may be more practical for strength coaches under time constraints (40 seconds of IRR required ~16 minutes to complete as compared with ~12 minutes with 20 seconds of IRR). Future research is needed to examine the longitudinal effects of IRR in training programs designed for the development of muscular power.

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